StrategiX 2.0 - Analytics Case Competition

Xavier School of Management (XLRI)

Team Name - 404 Not Found

"How do you save a subscriber who doesn't know they're leaving?"

Business Challenge :

5× more expensive	StreamMax detects churn only after it happens — re-engagement costs 5x more than prevention
? vacation vs. fatigue	The platform cannot distinguish between a user on vacation and a user genuinely drifting away
∞ lifetime value lost	Every undetected fatigued user represents permanently lost subscriber lifetime value
35.4% silently fatiguing	35.4% of users are silently fatiguing — undetected until it is too late

OUR ANALYTICAL APPROACH

01 — Problem Reframing
Shifted from reactive churn detection to proactive fatigue detection — catching users 30 days before cancellation

02 — Feature Engineering
Engineered 33 behavioural signals from raw usage data — trends, ratios, interactions across 10,000 users

03 — 4-Model Stacked Ensemble
XGBoost + LightGBM + Random Forest + CatBoost — blended by a Logistic Regression meta-learner

04 — Cross-Validated AUC : 0.790
5-Fold Stratified OOF validation — 79% chance of correctly ranking a fatigued user above an engaged one

10,000	33	4	0.790	35.4%	30
Users Analysed	Engineered Features	Models Stacked	OOF AUC (CV Estimate)	Fatigue Rate	Days Earlier Detection

KEY INSIGHTS

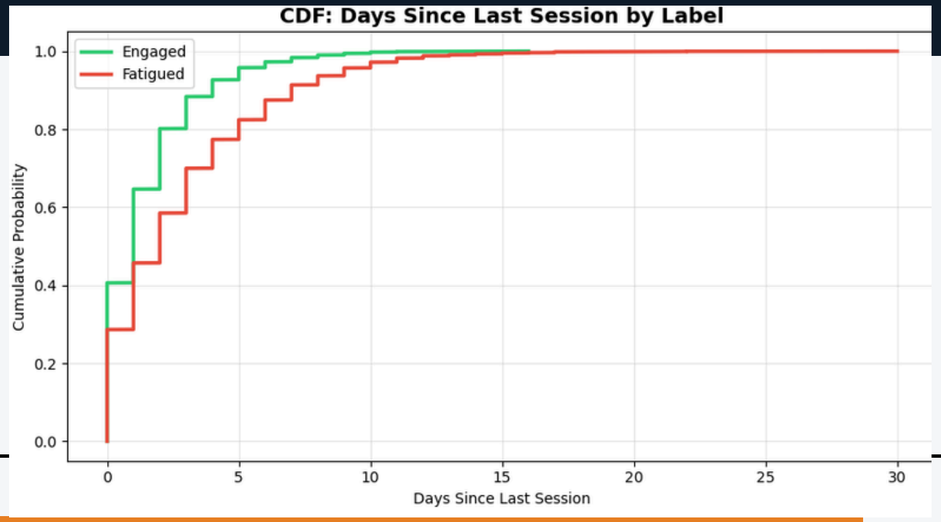
What Separates a Fatigued User From an Engaged One

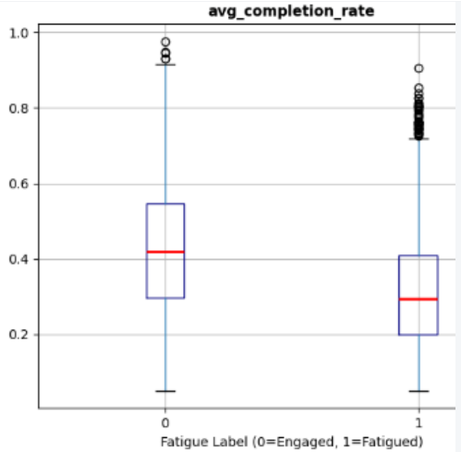
INSIGHT 01 — Recency Is the #1 Warning Signal

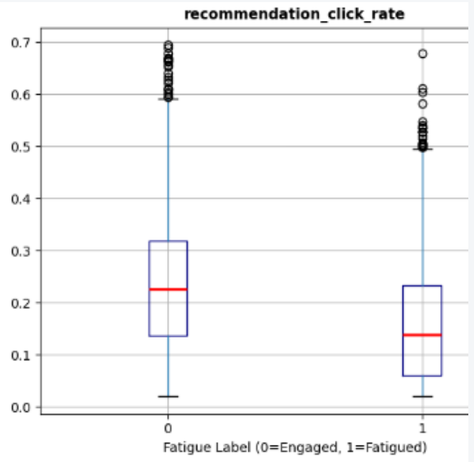
40%
Engaged users logged in TODAY

28%
Fatigued users logged in TODAY

Once a user crosses 7 days of inactivity the probability of fatigue jumps sharply — the habit is breaking







INSIGHT 02 — They Watch Less AND Enjoy It Less

Metric	Engaged	Fatigued	Drop
Completion Rate	0.425	0.315	-26%
Click Rate	0.231	0.157	-32%

Recommendation click rate drops from 0.231 to 0.157 — they have stopped trusting the platform to show them something worth watching

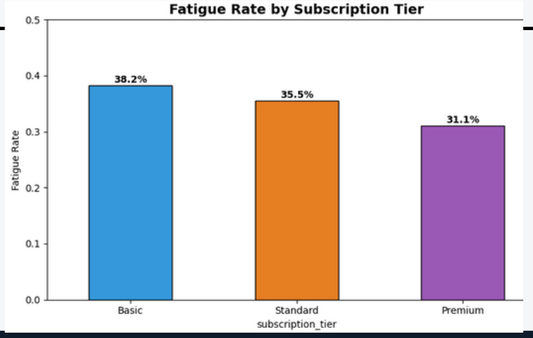
INSIGHT 03 — Basic Users & New Users Are Highest Risk

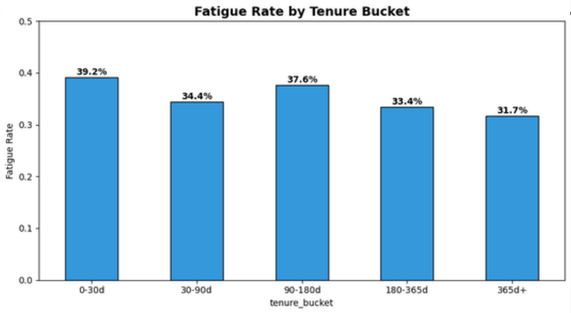
38.2%
Basic tier

35.5%
Standard tier

31.1%
Premium tier

Users in first 30 days churn at 39.2% — the onboarding window is the single





40% vs 28%	-26%	-32%	38.2%	39.2%	7 days
Login Today: Engaged vs Fatigued	Completion Rate Drop	Click Rate Drop	Basic Tier Fatigue Rate	New User (0-30d) Fatigue Rate	Inactivity Threshold

MODELLING APPROACH

How We Built the Early Warning System

DATA & FEATURE PIPELINE

RAW USAGE DATA
10,000 users · 12 behavioral signals

FEATURE ENGINEERING
33 signals: trends · ratios · interactions · flags

OUTLIER CAPPING
Winsorisation at 1st/99th percentile · train thresholds only

XGB	LGB	RF	CAT
Boosting Level-wise	Boosting Leaf-wise	Bagging Independ.	Ordered Boosting

LOGISTIC REGRESSION
Meta-Learner · learns optimal model weights

FATIGUE PROBABILITY
Per-user score 0.0 → 1.0 · 2,000 predictions

WHY THIS MODEL FAMILY

No feature scaling
Trees split on thresholds — our mixed-scale features work as-is

Non-linear patterns
Fatigue is not linear — a user with 0 sessions AND 10 days away is exponentially more at risk

Interpretable output
Feature importance + SHAP values translate directly to business action

Diverse error types
XGB, LGB, RF and CatBoost each make different mistakes — ensemble cancels them out

Neural nets rejected
8,000 rows too small — trees consistently outperform deep learning on structured tabular data

TOP SIGNALS IN BUSINESS TERMS

01 Days Since Login
Is the habit breaking?
corr: +0.26

02 Watch Time Trend
Are they watching less than usual?
corr: -0.30

03 Completion Rate
Are they enjoying the content?
corr: -0.29

04 Recommendation Clicks
Do they trust the platform?
corr: -0.28

05 Composite Fatigue Score
All risk signals in one number
corr: +0.43

8,000	33	4	5-Fold	0.790	2,000
Training Users	Engineered Features	Base Models	Stratified CV	OOF AUC (CV Estimate)	Test Predictions

BUSINESS INTERPRETATION

Who Is at Risk and Why

01 Silent Drifter

Standard / Basic · 90-180 days tenure

Sessions declining week over week. Still logging in but watching less and finishing less each week. Will cancel in 30-60 days.

MEDIUM-HIGH

~28% of risk

`minutes_trend_ratio < 0.7`

02 Lapsed Binge Watcher

Any tier · 60+ days tenure

Previously 3+ binges/month, now zero. Has lost emotional connection to the platform — hardest segment to win back.

VERY HIGH

~18% of risk

`binge_ratio dropped to 0`

03 New User Dropout

Any tier · 0-30 days tenure

39.2% fatigue rate — highest of all groups. Never found content that hooked them. Will cancel before the second billing cycle.

CRITICAL

~15% of risk

`is_new_user=1, completion<0.25`

04 Dissatisfied Viewer

Any tier · Any tenure

Completion rate <0.25, click rate <0.10. Keeps trying content but abandoning it — algorithm failing them, not platform fatigue.

HIGH

~20% of risk

`engagement_quality < 0.15`

05 Price-Sensitive Basic

Basic tier · 30-180 days tenure

Engagement declining but still active. Limited Basic tier content library is the trigger — comparing with competitor platforms.

MEDIUM

~19% of risk

`tier=Basic, fatigue 0.40-0.65`

KEY DISENGAGEMENT FACTORS — MODEL CONFIRMED

#1 Composite Fatigue Score
All risk signals in one number
+0.43

#2 Watch Time Trend (7d/30d)
Watching less than their own historical baseline
-0.30

#3 Content Completion Rate
Starting but not finishing — content mismatch
-0.29

#4 Recommendation Click Rate
Stopped trusting the algorithm to show good content
-0.28

#5 Days Since Last Session
Daily habit is breaking down
+0.26

5	39.2%	38.2%	+0.43	28%	-32%
User Segments Identified	New User Fatigue Rate	Basic Tier Fatigue Rate	Fatigue Score Correlation	Fatigued Users Login Today	Click Rate Drop in Fatigue

STRATEGIC RECOMMENDATIONS

Six Actions StreamMax Can Take Today

REC 01 — Fatigue Score Dashboard

Data + Retention

TRIGGER: Daily model run on all users

ACTION: Red (>0.65) / Amber (0.40-0.65) / Green (<0.40) alerts. Red users trigger automatic outreach within 24 hours

WHY: *Model catches at-risk users 30 days early — that window only has value if the team acts within it*

REC 02 — Personalised Re-engagement

Marketing + Algorithm

TRIGGER: 7d minutes dropped >40% vs 30d baseline

ACTION: Push notification or email — content based on user's top genre from watch history, not generic top-10 list

WHY: *Recommendation click rate was our 4th strongest signal — these users stopped trusting the algorithm*

REC 03 — Fix Onboarding (Day 0-30)

Product team

TRIGGER: All new users on signup

ACTION: Mandatory taste profile: 5 genre + 3 format preferences. Seed first 2 weeks of recommendations exclusively from this profile

WHY: *New users have the highest fatigue rate (39.2%) — they churn before finding their content identity on the platform*

REC 04 — Binge Recovery Campaign

Content + Marketing

TRIGGER: 3+ binges last month → 0 this month

ACTION: Continue your journey notification — reference last series watched, highlight new episodes or similar content in same genre

WHY: *Binge watchers have proven deep engagement. Drop to zero is content-driven not platform fatigue — one nudge restarts the habit*

REC 05 — Tier Upgrade Incentive

Growth + Pricing

TRIGGER: Basic tier user, fatigue score 0.40-0.65

ACTION: Offer 2 months Standard tier at 50% off — limited time, personalised push notification

WHY: *Basic tier churn 38.2% vs Premium 31.1% — a 7 point gap directly linked to content access limitations*

REC 06 — Recommendation Algorithm Reset

ML / Rec team

TRIGGER: Click rate below 0.10 for 2+ consecutive weeks

ACTION: Switch from collaborative filtering to genre-popularity recommendations within historically preferred genres

WHY: *Algorithm is stuck showing rejected content. A reset breaks the feedback loop and restores platform trust*

6	Day 0	24 hrs	50%	0.10	40%
Action Items	Onboarding Fix Starts	Red Alert Response Time	Tier Upgrade Discount	Click Rate Reset Trigger	Watch Time Drop Trigger

IMPLEMENTATION ROADMAP

From Model to Action — 90-Day Deployment Plan

01 DEPLOY
Week 1-2

- Deploy fatigue score pipeline on all users
- Build Red / Amber / Green retention dashboard
- Flag all current Red users (score >0.65)
- Immediate outreach to highest-risk users

→ Live risk score for every user

02 TEST
Month 1-2

- Launch personalised nudges for Silent Drifters
- Revised onboarding taste profile for new users
- A/B test: nudge group vs control group
- Measure causal lift from interventions

→ First data on true intervention ROI

03 SCALE
Month 2-3

- Tier upgrade campaign for Amber Basic users
- Binge Recovery campaign for Lapsed Bingers
- Fatigue score feeds recommendation algorithm
- All 6 recommendations fully operational

→ Full retention system running

04 MONITOR
Ongoing

- Retrain model monthly on fresh data
- Track KPI 1: Fatigue rate trend MoM
- Track KPI 2: Intervention conversion rate
- Track KPI 3: Revenue per user — nudged vs control

→ Quarterly model + strategy review

MODEL LIMITATIONS — WHAT WE KNOW AND ARE TRANSPARENT ABOUT

Limitation	What It Means	How We Address
OOF AUC 0.790 is a CV estimate	True test AUC pending competition evaluation — may differ	**Reported as estimate · validate post-submission**
21% of predictions are wrong	Model incorrectly flags or misses roughly 1 in 5 users	**Human review for high-value Premium users before action**
Correlation ≠ Causation	High fatigue score does not guarantee intervention will work	**A/B test every campaign to measure true causal lift**
Model needs retraining	User behaviour drifts with content library changes and seasons	**Monthly retraining pipeline built into Phase 4**

2 weeks — To First Deployment

6 recommendations — Interventions Planned

A/B — Testing Every Campaign

3 KPIs — Success Metrics Tracked

Monthly — Model Retraining Cycle

0.790 — OOF AUC (CV Estimate)